The recovered 17 November 1997 X9.62 draft

Date: 2026-10-23 UTC. **Question:** Does the original November 1997 working draft contain a rule that rejects the eleven otherwise qualifying seed candidates? **Result:** We recovered the full 103-page draft. Its stated prime-field curve-generation procedure gives no additional candidate predicate separating those eleven from the published curves. It already prints the seven published 192-, 239-, and 256-bit prime curves, including P-256. The actual generator's decision trace is still missing.

Primary source and exact provenance

The Internet Archive item hosts a PDF of the ANSI X9.62 working draft. Its cover is dated **17 November 1997** (PDF page 1). The 591,731-byte, 103-page PDF has SHA-256 `0c231d5176fcf0b239b44732213d281b518a0e16cc6f13f7bd4039ec99807b6e`. Its PDF metadata says Acrobat Distiller 3.01 created the file on 11 December 1997; the archive item says it was uploaded in 2017. Metadata and archive upload do not independently notarize when every page first circulated. The cover, procedure, and P-256 table page were visually inspected. source_check.json binds the source hash, exact sample seeds, page locations, and prior check record.

The table in §H.5, PDF pages 97–101 (printed pages 89–93), explicitly calls the prime-field entries *sample curves* for implementation checking. It contains P-192, prime192v2, prime192v3, prime239v1, prime239v2, prime239v3, and P-256. The seven 160-bit seeds in that table match the named published seeds exactly, verified by verify_source.py against the PDF text. In particular, PDF page 100 (printed page 92) gives P-256′s `C49D3608...` seed and `5AC635D8...` coefficient. This places those seven published parameter sets in a dated **November/December 1997 artifact**, much earlier than the September 1998 cover of the later draft. It does not say when the May and August generation runs ended or how competing candidates were evaluated.

What the draft says to accept

Section F.3.2, PDF page 63 (printed page 55), specifies: choose a seed and derive a curve, count its order, check nonsingularity, test near-primality of the order, check MOV and anomalous conditions, find a prime-order point, and output the curve. The note on PDF page 64 requires the order lower bound `r_min` to exceed `2^160`, advises `r_min <= p+1−2√p` for a prime field, and says the trial-division limit is typically 255. Section F.2.2, PDF page 62, defines near-primality through a smooth cofactor and a probably prime large factor. The **prime-field** seed instruction is §F.3.3.b on PDF page 65: choose an arbitrary seed bit string at least 160 bits long. The visible text gives **no schedule for the next seed after a candidate is rejected or passed** and no rule for selecting one of several passing curves for the published examples.

The existing exact 21-row workflow comparison records eleven omitted and ten published classes satisfying the named cofactor-one, `n>2^160`, nonanomalous, and stronger-than-required `MOV B=100` checks. verify_source.py hashes and checks that prior matrix; it does not rerun its arithmetic. Since every omitted curve has prime full order, its cofactor is 1 and its near-primality check succeeds for any ordinary trial-division bound. The verifier also checks each omitted curve's `p,n` against the Hasse interval. If the draft's advised `r_min` ceiling was followed, `r_min` cannot exclude any correctly counted prime-order curve over that field. An operator could choose a higher threshold, but the draft supplies no such value or candidate decision. Thus the **stated November 1997 procedure** does not explain passing over the eleven. A local software error, unpublished policy, different input or state, or later review remains possible. The stipulation that the historical program actually evaluated and rejected all eleven remains a hypothesis.

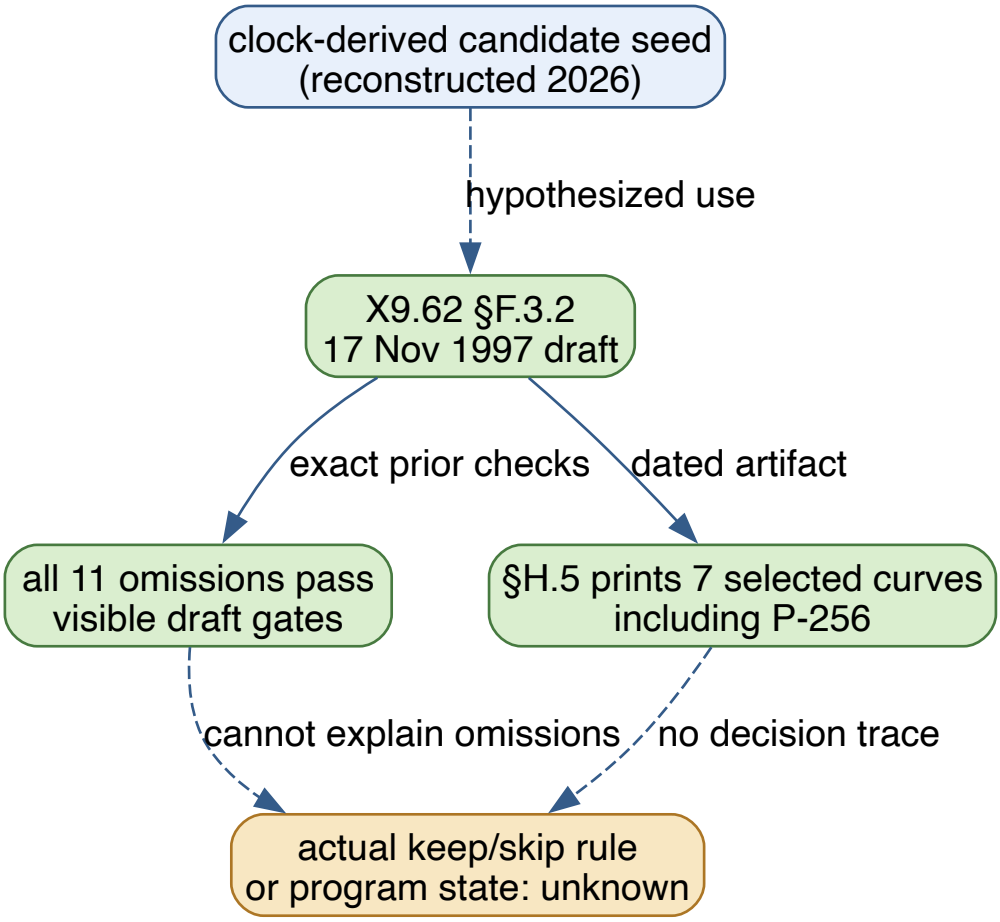


Figure 1: Draft procedure and unresolved decision

What changed, and what would settle the question

The missing November draft is now acquired. Its parameter appendix establishes that the seven published examples were fixed in this available artifact by its stated date and PDF creation timestamp. The earlier **21 June 1996**, **15 January 1997**, and **11 June 1997** drafts cited by independent sources remain acquisition targets, especially for how the procedure changed before and during the May seed runs. The original program, its run log, and candidate-level review notes would be stronger evidence for the actual keep/skip decisions.

An independently specified weakness predicate must still classify all eleven omissions and matched published controls, including P-256 blocks 876 and 1842, and yield a checkable attack mechanism or historical decision record. The recovered draft supplies neither. Its dated table strengthens provenance and rules out an extra **documented** November 1997 acceptance gate; it does not identify the postulated common weakness. The open selection problem remains open. No ECDLP cost claim or official research-status transition follows.

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Download the PDF from the source link and run `python3 research/nist_seed_1997_original_draft_20261023/verify_source.py --pdf /path/to/x962-11-17-97.pdf --output /tmp/source_check.json`. The script requires the exact SHA-256 and seven table seeds, and validates the previously archived gate matrix's identity and true predicates. The supplied source PDF is linked and hash-bound, not redistributed. The report, editable diagram, SVG and visually inspected PDF record this substantive source search. Source statements are `retrieved`; matched-curve screening is `internal` and `derived`; historical intent and execution are unobserved.